

MULTIMEDIA TECHNOLOGY

Unit-4: Video in Multimedia

Comprehensive Study Notes

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1. Basics of Video

Definition: Video is a technology that captures, records, processes, transmits, and reconstructs moving visual images. It is a sequence of still images (frames) displayed in rapid succession to create the illusion of motion, typically accompanied by audio.

Key Video Terminology

Fundamental Video Concepts

Term	Description
Frame	Single still image in a video sequence; the building block of video. Each frame differs slightly from the previous one.
Frame Rate (FPS)	Number of frames displayed per second. Determines motion smoothness. Higher FPS = smoother motion. 24 fps = Film, 30 fps = TV, 60 fps = Gaming.

Resolution	Number of pixels in each frame (Width × Height). Higher resolution = more detail. 1920 × 1080 = Full HD.
Aspect Ratio	Width to height ratio. 4:3 = Standard, 16:9 = Widescreen. Determines display shape.
Bit Rate	Amount of data processed per second. Affects quality and file size. Higher bit rate = better quality. Measured in Mbps (Megabits per second).
Compression	Reduces file size for storage/transmission. Lossy (loses quality) vs Lossless (preserves quality). Codec determines compression method.

Video Frame Rate Standards

Common Frame Rate Standards by Application

Standard	Frame Rate	Application
Film	24 fps	Cinema, Movies
PAL	25 fps	European TV (UK, Europe)
NTSC	30 fps	US TV, Japan, North America
HDTV	30/60 fps	High Definition TV
Web Video	30 fps	YouTube, Vimeo, Online
Gaming	60+ fps	Video Games, High-end
Slow Motion	120+ fps	Special effects, Sports
VR/AR	90+ fps	Virtual/Augmented Reality

Frame Rate Effects: 24 fps — Cinematic, film-like feel; 25 fps — European TV standard; 30 fps — US TV standard, smooth; 60 fps — Very smooth, good for fast motion; 120+ fps — Slow motion capability.

2. Analog and Digital Video

Analog Video

Definition: Video signal that uses continuous electrical signals to represent picture and sound.

Analog Video Characteristics

- Continuous signal
- Subject to signal degradation
- Quality degrades with copying
- Limited by bandwidth
- Vulnerable to interference
- No compression possible
- Older technology

Analog Video Formats

Consumer and Professional Analog Formats

Category	Formats
Consumer Formats	VHS (Video Home System) — Most popular; Betamax — Sony format, better quality; Video 8 — 8mm camcorder format
Professional Formats	Betacam — Professional broadcast; U-matic — Broadcast standard; 1-inch Type C — Studio format

Analog Signal Types

- **Composite Video** — All signals in one cable
- **S-Video (Y/C)** — Separate luminance & chroma
- **Component Video** — R, G, B separate signals
- **RF (Radio Frequency)** — Modulated signal

Digital Video

Definition: Video signal that uses discrete (binary) values to represent picture and sound.

Digital Video Characteristics

- Discrete (binary) signal
- No degradation with copying
- Can be compressed
- Resistant to interference
- Can be edited easily
- Supports metadata
- Modern technology

Digital Video Container Formats

Common Digital Video File Containers

Format	Extension	Description
AVI	.avi	Audio Video Interleave — Windows
MP4	.mp4	MPEG-4 — Universal standard
MOV	.mov	QuickTime — Apple
WMV	.wmv	Windows Media Video — Windows
MKV	.mkv	Matroska — Open source
FLV	.flv	Flash Video — Web (older)
WEBM	.webm	Modern web standard

Digital Video Parameters

- Resolution (width × height)
- Frame rate (fps)
- Bit rate (Mbps)
- Color depth (bits per pixel)

- Codec used (compression/decompression)
- Container format

Analog vs Digital Video Comparison

Key Differences Between Analog and Digital Video

Feature	Analog Video	Digital Video
Signal Type	Continuous signal	Discrete (binary) signal
Quality on Copying	Degrades with copying	No degradation
Compression	No compression	Can be compressed
Noise Resistance	Susceptible to noise	Noise-resistant
Editing	Difficult	Easy
Resolution	Limited	High resolution possible
Storage	Physical media	Digital media storage
Examples	VHS, Betacam	MP4, AVI, MOV
Broadcast	NTSC, PAL	HDTV, 4K, 8K

Conversion Process: Analog Video → ADC (Video Capture / Analog to Digital Conversion) → Digital Video

3. How to Use Video on PC

Video Software on PC

Video Players

Popular Video Player Software

Player	Description
VLC Media Player	Universal, supports all formats
Windows Media Player	Windows default
MPC-HC	Lightweight, efficient
KMPlayer	Feature-rich
PotPlayer	Modern, customizable

Video Editing Software

Categories of Video Editing Software

Category	Software
Professional	Adobe Premiere Pro, Final Cut Pro, DaVinci Resolve, Vegas Pro
Consumer / Semi-Professional	Adobe Premiere Elements, Corel VideoStudio, CyberLink PowerDirector, Pinnacle Studio
Free / Open Source	Shotcut, OpenShot, DaVinci Resolve (Free), HitFilm Express, Lightworks

Video Converters

Popular Video Conversion Tools

Tool	Description
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HandBrake	Free, powerful conversion
FFmpeg	Command-line tool
Any Video Converter	Easy to use
WinFF	GUI for FFmpeg

4. Introduction to Graphics Accelerator Cards

Graphics Accelerator Card (GPU — Graphics Processing Unit): A specialized processor designed to accelerate graphics rendering and video processing tasks.

Types of Graphics Cards

Graphics Card Categories

Type	Description	Examples
Integrated Graphics	Built into CPU; lower performance; uses system RAM; no additional cost; suitable for basic tasks, web browsing, office work	Intel HD Graphics, AMD Radeon (integrated)
Dedicated Graphics Card	Separate card in PCIe slot; high performance; has own VRAM; better power and cooling; suitable for gaming, video editing, 3D rendering	NVIDIA GeForce, AMD Radeon
Workstation Graphics Card	Professional grade; optimized for CAD, 3D, scientific; ECC memory support; certified for specific applications	NVIDIA Quadro, AMD Radeon Pro

Leading GPU Manufacturers

Manufacturer	Product Lines
NVIDIA	GeForce (Gaming), Quadro (Professional)
AMD (ATI)	Radeon (Gaming), Radeon Pro (Professional)
Intel	Iris Xe, Arc (New)

GPU Specifications

Key GPU Technical Specifications

Specification	Description
CUDA Cores (NVIDIA)	Parallel processing units
Stream Processors (AMD)	AMD's parallel processing units
VRAM	Video memory (GB)
Memory Type	GDDR5, GDDR6, HBM
Memory Bus Width	128-bit, 256-bit, 384-bit
Core Clock Speed	MHz — Processing speed
Memory Clock Speed	MHz — Memory speed
TDP (Thermal Design Power)	Power consumption (Watts)
DirectX Support	Graphics API support
OpenGL Support	Graphics API support
Ray Tracing Support	Realistic lighting support

Benefits of Graphics Card for Multimedia

GPU Benefits Across Multimedia Applications

Application	Benefits
Video Editing	Faster rendering, real-time preview, support for higher resolutions (4K, 8K), GPU acceleration for effects
3D Animation & Rendering	Fast 3D rendering, real-time viewport updates, support for complex scenes, ray tracing capability
Gaming	High frame rates, high-quality graphics, VR support, ray tracing for realistic lighting
Video Playback	Smooth 4K/8K playback, hardware decoding, color accurate display
Graphic Design	Smooth UI experience, large canvas support, multi-monitor support

5. Brief Note on Various Video Standards

5.1 NTSC (National Television System Committee)

NTSC Standard Specifications

Property	Value
Full Name	National Television System Committee
Region	North America, Japan, Philippines, Taiwan
Type	Analog Color Television System
Frame Rate	30 fps (29.97 actual)
Lines Per Frame	525 (visible: 480)
Aspect Ratio	4:3
Color Subcarrier	3.58 MHz
Scan Type	Interlaced
Field Rate	60 fields/sec

Characteristics: Older analog standard. Also known as "Never The Same Color" due to color issues. 480i resolution. Being replaced by digital standards (ATSC). Used for DVD-Video in NTSC regions.

5.2 PAL (Phase Alternating Line)

PAL Standard Specifications

Property	Value
Full Name	Phase Alternating Line

Region	Europe, UK, Australia, India, China, Africa
Type	Analog Color Television System
Frame Rate	25 fps
Lines Per Frame	625 (visible: 576)
Aspect Ratio	4:3 (originally)
Color Subcarrier	4.43 MHz
Scan Type	Interlaced
Field Rate	50 fields/sec

Characteristics: Better color stability than NTSC. 576i resolution. Variants: PAL-I, PAL-B/G, PAL-D/K, PAL-M.

5.3 SECAM («Séquentiel Couleur À Mémoire»)

SECAM Standard Specifications

Property	Value
Full Name	Séquentiel Couleur À Mémoire
Region	France, Russia, Eastern Europe, Africa
Type	Analog Color Television System
Frame Rate	25 fps
Lines Per Frame	625
Aspect Ratio	4:3 (originally)
Color Transmission	Sequential
Scan Type	Interlaced
Color System	FM for color

Characteristics: Different color encoding method. Sequential color memory. Resistant to phase errors. Less common worldwide. Being replaced by digital standards.

Analog Standard Comparison Table

NTSC vs PAL vs SECAM

Feature	NTSC	PAL	SECAM	Description
Frame Rate	30 fps (29.97)	25 fps	25 fps	Motion smoothness
Lines	525	625	625	Vertical resolution
Visible Lines	480	576	576	Active picture area
Aspect Ratio	4:3	4:3	4:3	Width:Height ratio
Color System	3.58 MHz	4.43 MHz	FM	Color encoding
Color Stability	Good	Better	Best	Color consistency
Regions	USA, Japan	Europe, UK, India	France, Russia	Geographic coverage

5.4 HDTV (High Definition Television)

HDTV Standard Overview

Property	Details
Full Name	High Definition Television
Type	Digital High Resolution Television Standard
Introduction	1990s (Japan), 2000s (Worldwide)

HDTV Resolutions

HDTV Resolution Variants

Standard	Resolution	Aspect Ratio	Scan
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720p (HD Ready)	1280 × 720	16:9	Progressive
1080i (Full HD)	1920 × 1080	16:9	Interlaced
1080p (Full HD)	1920 × 1080	16:9	Progressive

Advantages Over SD (Standard Definition)

- 5–6 times more detail than SD
- Widescreen 16:9 aspect ratio
- Digital signal (no interference)
- Dolby Digital surround sound
- Multiple audio/subtitle streams
- Electronic Program Guide (EPG)

Resolution Standards (Progression)

From SD to 8K

Standard	Resolution (pixels)
SD (Standard Definition)	720 × 576 (PAL) / 720 × 480 (NTSC)
HD Ready (720p)	1280 × 720
Full HD (1080p)	1920 × 1080
Quad HD (2K)	2560 × 1440
Ultra HD (4K)	3840 × 2160
DCI 4K (Cinema)	4096 × 2160
8K UHD	7680 × 4320

Progressive vs Interlaced Scan: **Interlaced (i)** draws odd lines then even lines (two fields per frame). **Progressive (p)** draws all lines sequentially (one field per frame). 1080i: 60 fields/sec, 30 frames/sec (interlaced). 1080p: 60 frames/sec (progressive) — smoother.

6. Introduction to Video Capturing Media & Instruments

Video Capturing Instruments

Video Capture Hardware Types

Device	Description	Examples
Video Capture Card	Installed in PC; PCIe/USB connection; captures analog/digital video; converts to digital format	Elgato, Blackmagic, AverMedia
Capture Device / Interface	External device; USB/Thunderbolt connection; supports various inputs; portable	HDMI capture devices
TV Tuner Card	Captures TV broadcast; analog/digital tuner; antenna/cable input; records TV shows	—
DVR (Digital Video Recorder)	Standalone device; records to hard drive; security systems; time-shift viewing	—
Video Digitizer	Converts analog to digital; VHS to digital conversion; USB devices	—

Video Capture Specifications

Key Specifications for Capture Devices

Specification	Importance
Resolution Support	Max resolution that can capture
Frame Rate Support	Max fps that can capture
Input Interfaces	HDMI, Composite, Component, S-Video, SDI
Output Format	MP4, AVI, MOV, etc.

Bit Rate	Capture quality/size
Hardware Encoding	On-board compression
Latency	Real-time capture capability
Audio Support	Embedded/separate audio
Compatibility	OS and software support

7. Videodisc

Definition: Videodisc is an optical disc storage medium used to store video and audio content. It predates DVD and was available in various formats. Introduced in the 1970s. Predecessor of DVD and Blu-ray.

Videodisc Formats

Three Major Videodisc Formats

Format	Year	Key Characteristics
LaserDisc	1978	12-inch disc; analog video; digital audio; 30 min per side; CAV/CLV formats
CED (Capacitance Electronic Disc)	1981	RCA format; grooved disc; capacitance pickup; 60 min per side
VHD (Video High Density)	1983	JVC format; Japanese market

LaserDisc Features

CAV vs CLV on LaserDisc

Feature	CAV (Constant Angular Velocity)	CLV (Constant Linear Velocity)
Capacity	30 min per side	60 min per side
Still Frame	Yes	No
Random Access	Yes	No (sequential only)
Frames per Side	54,000	Higher capacity

8. Virtual Reality Terminology

Definition: Virtual Reality is a computer-generated simulation of a 3D environment that users can interact with using special equipment.

Types of VR

VR Immersion Levels

Type	Description	Examples
Non-Immersive VR	Computer screen display; 3D content on 2D screen	3D games on monitor
Semi-Immersive VR	Large screen/projection; partial immersion	Flight simulators
Fully Immersive VR	HMD (Head Mounted Display); complete immersion	Oculus, HTC Vive

8.1 HMD (Head Mounted Display)

Definition: Display device worn on the head that provides visual information to the user.

HMD Components

- Display(s) (OLED, LCD)
- Lenses (Optics)
- Head tracking sensors (gyroscope, accelerometer)
- Audio (headphones/speakers)
- Microphone (for voice input)
- Eye tracking (in advanced models)
- Connectivity (USB, HDMI, Wireless)

Types of HMD

HMD Categories

Type	Description	Examples
PC-VR HMD	Connected to PC; high performance	Oculus Rift, HTC Vive, Valve Index
Standalone HMD	All-in-one; no external device needed	Oculus Quest, HTC Vive Focus
Smartphone-Based HMD	Uses smartphone as display; affordable	Google Cardboard, Samsung Gear

8.2 BOOM (Binocular Omni-Orientation Monitor)

Definition: A counter-balanced arm-mounted display system for VR applications.

BOOM Characteristics

- Mounted on articulated arm
- Counterbalanced for weightlessness
- Handles heavy displays
- Head tracking through arm position
- No weight on head
- Physical movement required

BOOM Features

- Better for heavy/expensive displays
- Good for professional applications
- Less comfortable for long use
- Limited movement range
- Not common for consumer use

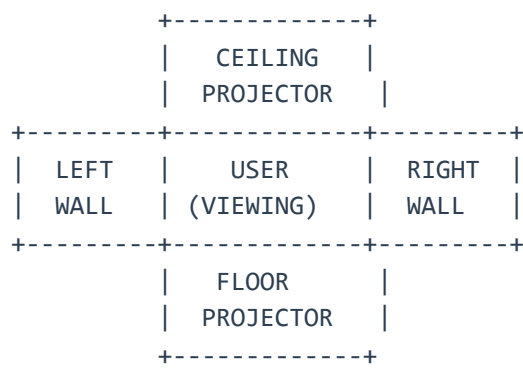
8.3 CAVE (Cave Automatic Virtual Environment)

Definition: A room-sized VR environment with projection on walls, floor, and ceiling.

CAVE Characteristics

- Multiple projection surfaces
- Immersive room environment
- Stereoscopic projection
- Head tracking via markers
- Users wear lightweight glasses
- Group experience possible

Typical CAVE Setup



Applications of CAVE

- Scientific visualization
- Architectural walkthroughs
- Training simulations
- Military applications
- Research laboratories

8.4 Input Devices for VR

VR Input Technologies

Device	Description	Examples
Motion Controllers	Hand-held controllers; track hand movements; haptic feedback	Oculus Touch, HTC Vive Controllers

Data Gloves	Wearable gloves; track finger movements; pressure sensors; gesture recognition	—
Tracking Systems	Infrared sensors; inside-out tracking; external base stations	Lighthouse, Constellation
Eye Tracking	Tracks eye movements; foveated rendering; gaze-based interaction	—
Hand Tracking	Camera-based; no controllers needed; gesture recognition	Leap Motion

8.5 Sensual Technology (Sensory Feedback)

Definition: Technology that provides sensory feedback to enhance the VR experience beyond visual/audio.

Types of Sensory Feedback in VR

Type	Description	Examples
Haptic Feedback	Vibration/touch sensations; force feedback; texture simulation	Controller rumble, haptic gloves
Olfactory (Smell)	Scent generation; odor dispensing; enhances realism	—
Gustatory (Taste)	Taste simulation; emerging technology; limited applications	—
Thermal Feedback	Temperature simulation; heating/cooling; realistic sensations	—
Motion Platforms	Moving platforms; force feedback; motion simulation	VR treadmills, motion chairs

Quick Reference Summary

Key Terms

Essential Multimedia Terminology

Term	Definition
Frame	Single still image in video sequence
FPS (Frames Per Second)	Speed of video playback
Resolution	Number of pixels in each frame
Aspect Ratio	Width to height ratio (4:3, 16:9)
Bit Rate	Data processed per second (Mbps)
Codec	Compression/decompression algorithm
Container	File format that holds video/audio
GPU	Graphics Processing Unit
NTSC	US/Japan TV standard (30 fps)
PAL	European TV standard (25 fps)
HDTV	High Definition Television
HMD	Head Mounted Display
CAVE	Cave Automatic Virtual Environment
VR	Virtual Reality
Interlaced	Two fields per frame
Progressive	One field per frame

Video Standards Quick Reference

NTSC: 30 fps, 525 lines (480 visible), 4:3, US/Japan
PAL: 25 fps, 625 lines (576 visible), 4:3, Europe/India
SECAM: 25 fps, 625 lines, France/Russia
HDTV: 720p, 1080i, 1080p, 16:9

Resolution Standards

SD (Standard Def): 720 x 480 (NTSC), 720 x 576 (PAL)
HD Ready (720p): 1280 x 720
Full HD (1080p): 1920 x 1080
2K (Quad HD): 2560 x 1440
4K (Ultra HD): 3840 x 2160
8K (Ultra HD): 7680 x 4320

Video Format Recommendation Guide

Best Practices for Video Formats

Use Case	Container	Codec	Bitrate
YouTube Upload	MP4	H.264	8–12 Mbps (1080p)
Social Media	MP4	H.264	4–6 Mbps (720p)
Professional Edit	MOV/MXF	ProRes	50+ Mbps
DVD Authoring	VOB	MPEG-2	9 Mbps
Blu-ray Authoring	M2TS	H.264	20–40 Mbps
Streaming	MP4/WebM	H.265/VP9	4–8 Mbps (1080p)
Archiving	MKV/MP4	H.265	High quality